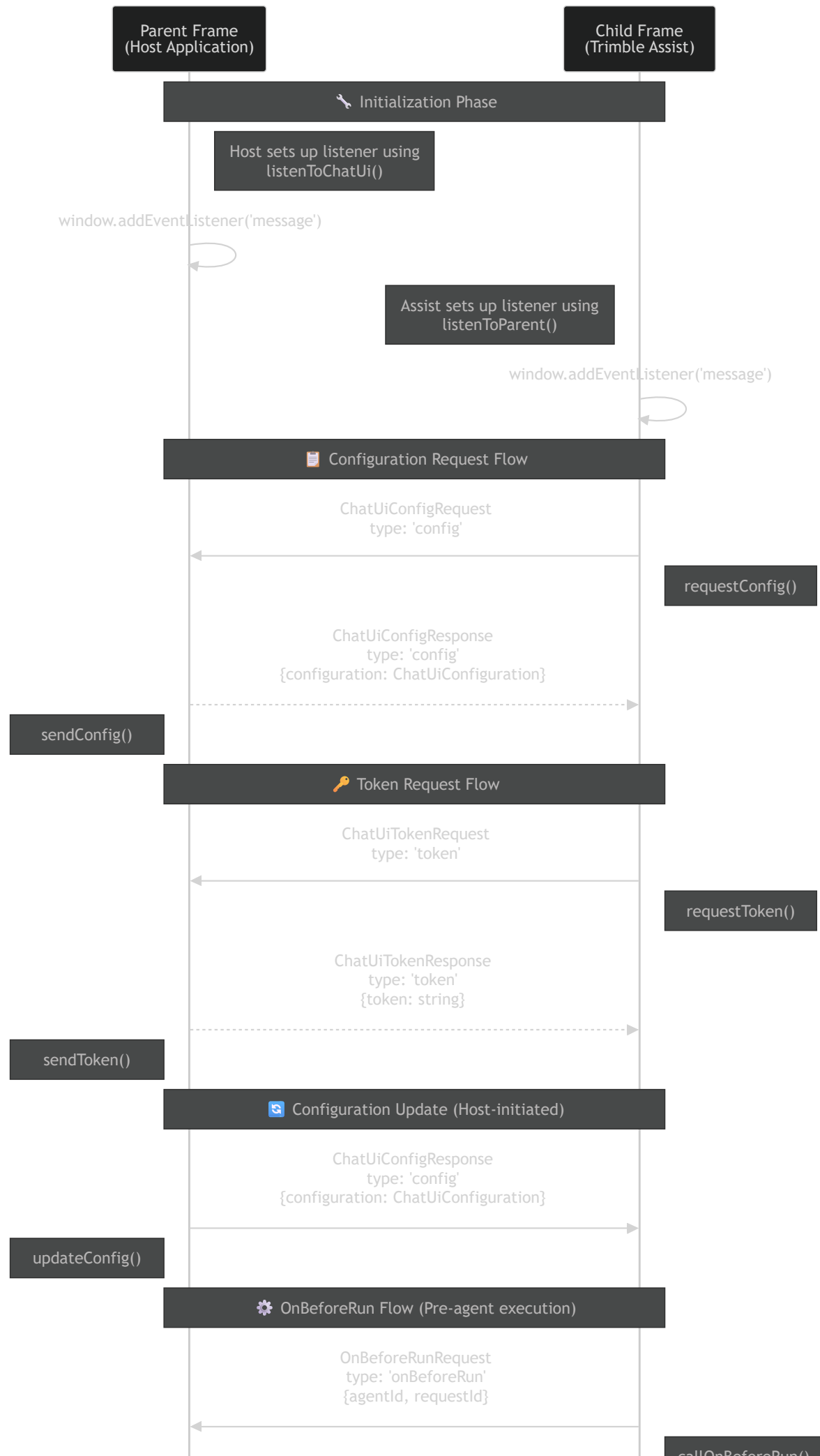


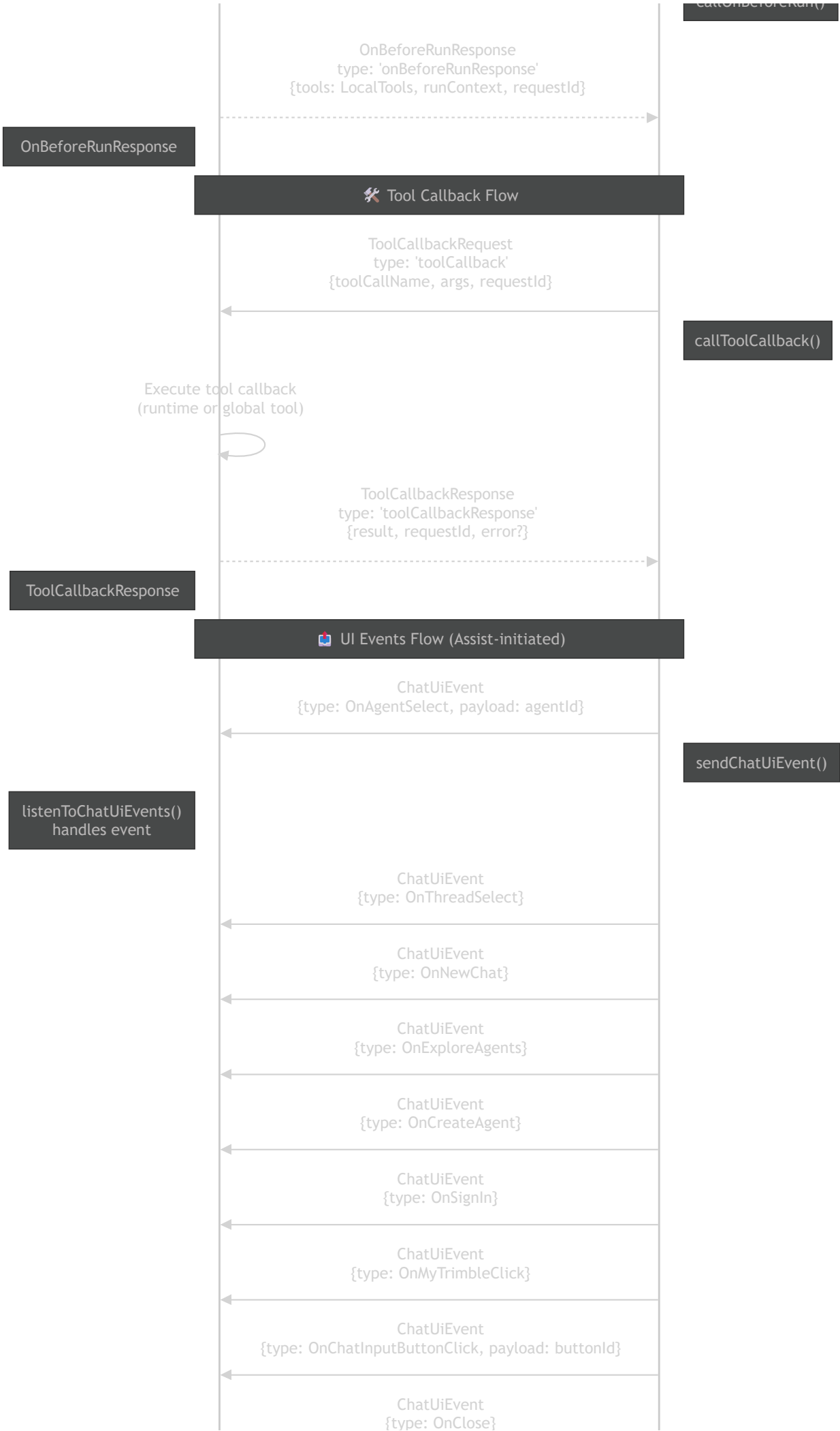
Iframe Communication Flow

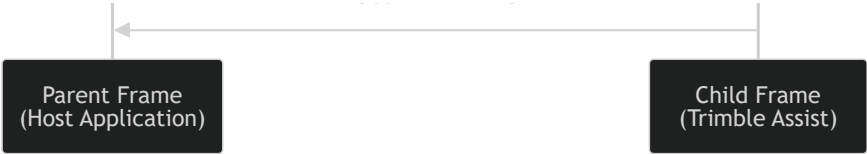
This document illustrates the communication flow between the **Parent Frame (Host Application)** and the **Child Frame (Trimble Assist)** using the `postMessage` API.

The SDK enables bi-directional communication between a host application and an embedded Trimble Assist iframe. Communication is achieved through the browser's `window.postMessage()` API with structured request/response patterns.

Sequence Diagram







Message Types

Requests (Child → Parent)

Request Type	Class	Description
config	ChatUiConfigRequest	Request the UI configuration from the host
token	ChatUiTokenRequest	Request the authentication token from the host
toolCallback	ToolCallbackRequest	Execute a tool callback on the host
onBeforeRun	OnBeforeRunRequest	Request tools and context before running an agent
onUnauthorized	OnUnauthorizedRequest	Notify parent that token is unauthorized and request handling

Responses (Parent → Child)

Response Type	Class	Description
config	ChatUiConfigResponse	Configuration response containing ChatUiConfiguration
token	ChatUiTokenResponse	Token response containing the auth token
toolCallbackResponse	ToolCallbackResponse	Result of a tool callback execution

Response Type	Class	Description
<code>onBeforeRunResponse</code>	<code>OnBeforeRunResponse</code>	Tools and context for agent execution
<code>onUnauthorizedResponse</code>	<code>OnUnauthorizedResponse</code>	Optional unauthorized handling message from parent

UI Events (Child → Parent)

Event Type	Payload	Description
<code>OnAgentSelect</code>	<code>string</code> (agentId)	User selected an agent
<code>OnThreadSelect</code>	<code>undefined</code>	User selected a thread
<code>OnNewChat</code>	<code>undefined</code>	User started a new chat
<code>OnExploreAgents</code>	<code>undefined</code>	User clicked explore agents
<code>OnCreateAgent</code>	<code>undefined</code>	User clicked create agent
<code>OnSignIn</code>	<code>undefined</code>	User clicked sign in
<code>OnMyTrimbleClick</code>	<code>undefined</code>	User clicked My Trimble
<code>OnChatInputButtonClick</code>	<code>string</code> (buttonId)	User clicked a custom input button
<code>OnClose</code>	<code>undefined</code>	User closed the Chat UI

SDK Methods

Parent Frame Helpers (`parent-frame.helpers.ts`)

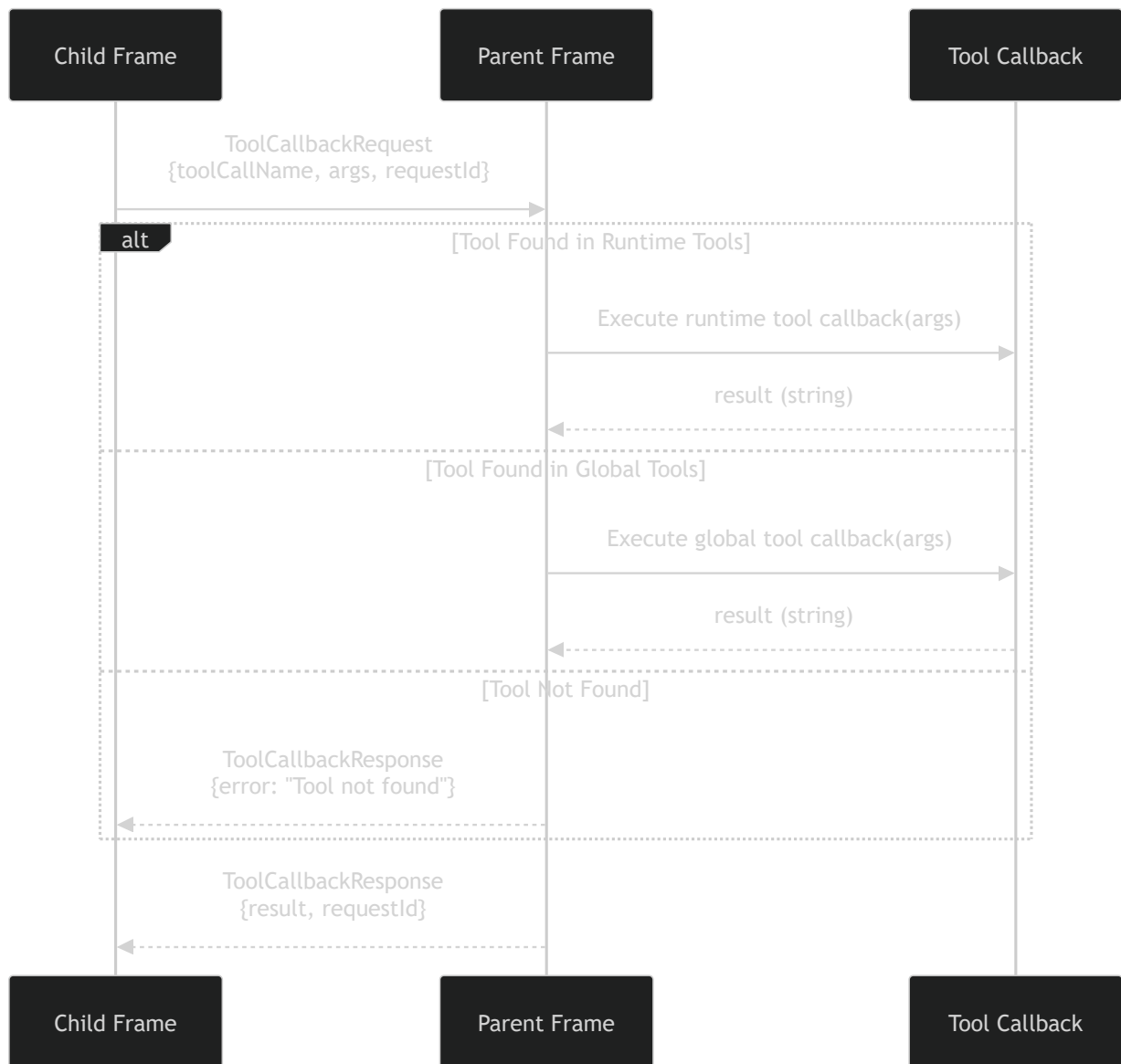
Method	Description
<code>listenToChatUi()</code>	Sets up message listener for config, token, tool callback, onBeforeRun, and onUnauthorized requests
<code>updateConfig()</code>	Sends a configuration update to the child frame
<code>listenToChatUiEvents()</code>	Sets up listener for UI events from the child frame
<code>subscribeToChatEvents()</code>	Sets up listener for internal chat lifecycle events (run, thread, agent, model)

Child Frame Helpers (`child-frame.helpers.ts`)

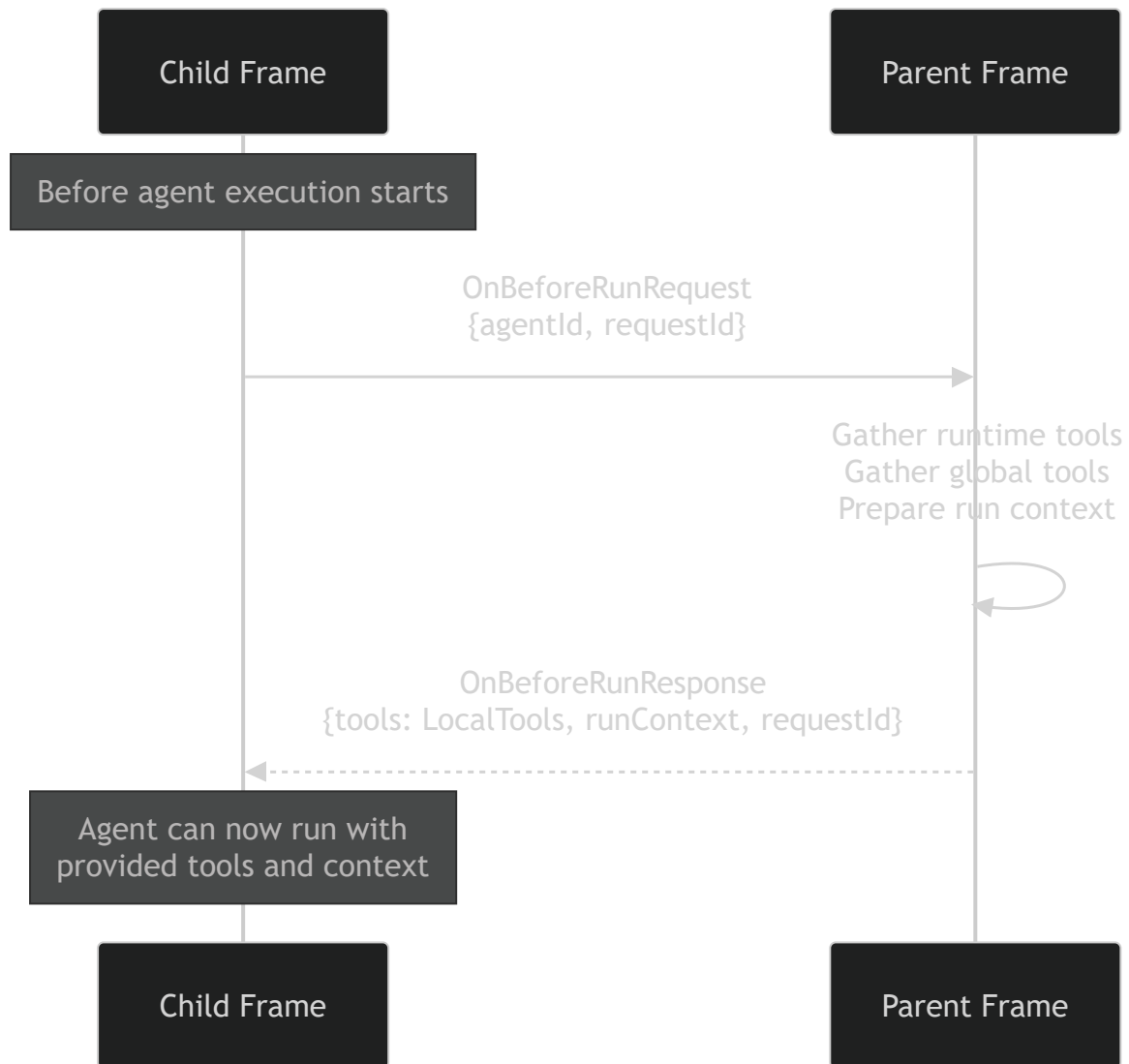
Method	Description
<code>requestConfig()</code>	Requests configuration from the parent frame
<code>requestToken()</code>	Requests authentication token from the parent frame
<code>onUnauthorized()</code>	Requests unauthorized handling response from parent
<code>callToolCallback()</code>	Executes a tool callback on the parent and awaits result
<code>callOnBeforeRun()</code>	Requests tools and context before running an agent
<code>sendChatUiEvent()</code>	Sends a UI event to the parent frame
<code>sendChatEvent()</code>	Sends an internal chat lifecycle event to the parent frame
<code>listenToParent()</code>	Sets up message listener for responses from parent

Data Flow Details

Tool Callback Flow



OnBeforeRun Flow



Configuration Structure

```

interface ChatUiConfiguration {
  environment: "development" | "stage" | "prod";
  onBeforeRunTimeout?: number;
  uiConfig: {
    theme: "dark" | "light";
    contentVariant: ContentVariants;
    variant: ChatUiVariants;
    chatInput: {
      buttons: ChatInputButton[];
      hideModelSelection: boolean;
    };
    showSignIn?: boolean;
    productAgents?: string[];
  };
  localization: {
    locale?: string; // Preferred - BCP-47 locale tag
  };
}
  
```

```
translations?: Translations; // Deprecated - use locale instead
selectedLanguage?: string; // Deprecated - use locale instead
};
agentId: string;
threadId?: string;
}
```

LocalTools Structure

```
interface LocalTools {
  runTime: {
    [toolCallId: string]: {
      definition: Tool;
      callback: ToolCallCallback;
      timeOutInMs?: number;
    };
  };
  global: {
    [toolCallId: string]: {
      callback: ToolCallCallback;
      timeOutInMs?: number;
    };
  };
}
```